

Physical Chemistry Practical 1: Volumetric Analysis – Acid-Base & Back Titrations

Investigation of Monoprotic, Diprotic, and Back Titration Techniques with Purity Analysis

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Physical Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Determination of the Concentration of a Monoprotic Acid (FA 2) by Titration

Cambridge Syllabus Reference: 9701/31/M/J/23/Q1 · [16 Marks]

Chemicals and Solutions Provided:

- FA 1 is aqueous sodium hydroxide, containing 4.20 g dm^{-3} of NaOH ($M_r = 40.0$).
- FA 2 is dilute hydrochloric acid, HCl, of unknown concentration.
- Methyl orange indicator solution.

Hazard & Safety Advice:

- Wear eye protection. FA 1 is corrosive and an irritant; FA 2 is an irritant. Wash splashes immediately.

Experimental Procedure:

1. Fill the burette with FA 2.
2. Pipette 25.0 cm^3 of FA 1 into a clean conical flask.
3. Add 2–3 drops of methyl orange indicator. The solution turns yellow.
4. Titrate FA 1 with FA 2 until the indicator turns permanently from yellow to orange/red.
5. Record your burette readings in the table below. Perform sufficient titrations to obtain concordant results (within 0.10 cm^3).

(a) Record your burette readings and the volumes of FA 2 added in the table below. Indicate your concordant titres with a tick (✓). [5]

Burette Readings	Rough	Titration 1	Titration 2	Titration 3
Final burette reading / cm^3				
Initial burette reading / cm^3				
Titre / cm^3				
Best titration results (✓)				

(b) From your selected titration results, calculate the mean volume (titre) of FA 2 required. Show which values you used. [1]

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(c) Calculate the amount, in moles, of sodium hydroxide present in the 25.0 cm^3 portion of FA 1 used in each titration. [2]

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(d) Write the balanced chemical equation for the neutralization reaction between HCl and NaOH. Hence deduce the amount, in moles, of HCl present in your mean titre of FA 2. [2]

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(e) Calculate the concentration of hydrochloric acid in FA 2, in mol dm^{-3} . Give your final answer to three significant figures. [2]

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(f) The manufacturer specifies that the uncertainty of each burette reading is $\pm 0.05 \text{ cm}^3$. Calculate the maximum percentage apparatus uncertainty in your titre volume. [2]

(g) A student washed the conical flask with the sodium hydroxide solution FA 1 immediately before pipetting the 25.0 cm^3 sample into it. Explain what effect, if any, this error would have on the calculated concentration of FA 2. [2]

Question 2: Back Titration: Determination of Calcium Carbonate Purity in Limestone**Cambridge Syllabus Reference: 9701/33/M/J/23/Q1 · [24 Marks]**

Chemicals and Solutions Provided:

- FA 3 is an impure solid limestone sample containing calcium carbonate, CaCO_3 .
- FA 4 is 1.00 mol dm^{-3} hydrochloric acid, HCl .
- FA 5 is $0.100 \text{ mol dm}^{-3}$ sodium hydroxide, NaOH .
- Phenolphthalein indicator.

Experimental Procedure:

1. Weigh accurately 1.50 g of the solid limestone sample FA 3 into a 250 cm^3 beaker.
2. Using a pipette, add exactly 50.0 cm^3 of 1.00 mol dm^{-3} HCl (FA 4) to the beaker. Effervescence occurs.
3. Swirl the beaker until all calcium carbonate has reacted and effervescence ceases completely.
4. Transfer the mixture quantitatively into a 250 cm^3 volumetric flask, rinse the beaker with distilled water, add the washings, and make up to the mark with distilled water. Invert thoroughly to mix. Label this solution FA 6.
5. Pipette 25.0 cm^3 of FA 6 into a conical flask, add 3 drops of phenolphthalein, and titrate with $0.100 \text{ mol dm}^{-3}$ NaOH (FA 5) from the burette.

(a) Record your titration results for the titration of 25.0 cm^3 portions of FA 6 with $0.100 \text{ mol dm}^{-3}$ NaOH (FA 5) in the table below. [5]

Burette Readings	Rough	Titration 1	Titration 2	Titration 3
Final burette reading / cm^3				
Initial burette reading / cm^3				
Titre / cm^3				
Best titration results ()				

(b) State the suitable mean titre of FA 5 to be used in your calculations. [1]

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(c) Calculate the amount, in moles, of NaOH present in your mean titre of FA 5. [2]

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(d) Deduce the amount, in moles, of unreacted HCl present in the 25.0 cm^3 sample of FA 6, and hence calculate the total amount of unreacted HCl in the entire 250 cm^3 volumetric flask. [3]

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(e) Calculate the initial amount, in moles, of HCl added to the limestone sample (in the 50.0 cm^3 of FA 4). [2]

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(f) Calculate the amount, in moles, of HCl that reacted with the calcium carbonate in the limestone sample. [2]

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(g) Write the balanced equation for the reaction of calcium carbonate with hydrochloric acid. Hence calculate the mass of calcium carbonate present in the 1.50 g limestone sample. [Ar: Ca = 40.1, C = 12.0, O = 16.0] [3]

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(h) Calculate the percentage by mass of calcium carbonate in the limestone sample. [2]

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(i) Suggest why back-titration is preferred over direct titration of solid calcium carbonate with hydrochloric acid. [2]

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(j) State how the student could verify that all the calcium carbonate in the limestone had completely dissolved in the acid before making up the solution to the mark. [2]

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical 1: Volumetric Analysis — Acid-Base & Back Titrations

Question 1: Determination of the Concentration of a Monoprotic Acid (FA 2) by Titration [16 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Award 1 mark for clear burette table with correct headers and units. Award 1 mark for recording all volumes to 0.05 cm ³ . Award 1 mark for obtaining two concordant titres within 0.10 cm ³ . Award 1-2 marks for accuracy compared to supervisor's titre (within 0.20 cm ³ = 2 marks; within 0.30 cm ³ = 1 mark).	[5]
(b)	Mean correctly calculated from selected ticked titres within 0.10 cm ³ , quoted to 2 decimal places (e.g. 23.45 cm ³).	[1]
(c)	Concentration of NaOH = $4.20 / 40.0 = 0.105 \text{ mol dm}^{-3}$ [1] Moles in 25.0 cm ³ = $0.105 \times (25.0 / 1000) = 2.625 \times 10^{-3} \text{ mol}$ [1]	[2]
(d)	HCl + NaOH → NaCl + H ₂ O [1] Mole ratio is 1:1, so moles of HCl = $2.625 \times 10^{-3} \text{ mol}$ [1]	[2]
(e)	Concentration of FA 2 = $(2.625 \times 10^{-3} / \text{Titre}) \times 1000$. For titre = 23.45 cm ³ , conc = $0.112 \text{ mol dm}^{-3}$. (Allow ECF from part (b)). [2]	[2]
(f)	Two readings per titre, total uncertainty = $2 \times 0.05 = 0.10 \text{ cm}^3$. % uncertainty = $(0.10 / \text{Titre}) \times 100\%$. (e.g. $0.10 / 23.45 \times 100\% = 0.426\%$). [2]	[2]
(g)	Droplets of FA 1 adhering to the flask walls increase the total moles of NaOH present. This requires a larger volume of FA 2 to reach the endpoint. Because Titre is in the denominator, the calculated concentration of FA 2 would be lower than the true value. [2]	[2]

Question 2: Back Titration: Determination of Calcium Carbonate Purity in Limestone [24 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Burette table formatted correctly, readings to 0.05 cm ³ , concordant titres within 0.10 cm ³ , accuracy marks [5].	[5]
(b)	Correct mean titre calculated from concordant runs to 2 decimal places (e.g. 21.80 cm ³).	[1]
(c)	Moles of NaOH = $0.100 \times (\text{Titre} / 1000)$. For 21.80 cm ³ : moles = $2.18 \times 10^{-3} \text{ mol}$. [2]	[2]
(d)	HCl + NaOH → NaCl + H ₂ O (1:1 ratio), so unreacted HCl in 25.0 cm ³ = $2.18 \times 10^{-3} \text{ mol}$ [1] Total unreacted HCl in 250 cm ³ = $2.18 \times 10^{-3} \times (250 / 25.0) = 2.18 \times 10^{-2} \text{ mol}$ (0.0218 mol) [2]	[3]
(e)	Initial moles of HCl = $1.00 \text{ mol dm}^{-3} \times (50.0 / 1000) = 0.0500 \text{ mol}$. [2]	[2]
(f)	Reacted moles of HCl = Initial moles - Unreacted moles = $0.0500 - 0.0218 = 0.0282 \text{ mol}$. [2]	[2]
(g)	CaCO ₃ + 2HCl → CaCl ₂ + H ₂ O + CO ₂ [1] Moles of CaCO ₃ = Moles of HCl reacted / 2 = $0.0282 / 2 = 0.0141 \text{ mol}$ [1] Mr(CaCO ₃) = 40.1 + 12.0 + 48.0 = 100.1 g/mol. Mass of CaCO ₃ = $0.0141 \times 100.1 = 1.411 \text{ g}$ [1]	[3]
(h)	% purity = $(1.411 \text{ g} / 1.50 \text{ g}) \times 100\% = 94.1\%$. [2]	[2]
(i)	Calcium carbonate is an insoluble solid that reacts slowly with acid near the endpoint; determining a sharp visual indicator endpoint with a solid suspension is difficult and inaccurate. Dissolving completely in excess acid produces a homogeneous solution that allows a sharp, precise endpoint with standard alkali. [2]	[2]
(j)	Ensure effervescence has completely stopped and no solid particles remain at the bottom of the beaker (or only insoluble siliceous grit/sand remains). Gentle warming can be applied to ensure complete reaction. [2]	[2]